

Laura Risley

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Education

PhD in Mathematics, University of Reading

2021-Present

CASE studentship with the Met Office UK

Planned submission date: July 2025

'Assimilation of future ocean-current measurements from satellites'

I am adapting the velocity variables used in NEMOVAR, a variational data assimilation software. This involves coding an idealised model (Shallow Water Equations); theoretical and numerical analysis and running experiments with the NEMO model on Monsoon.

MSc modules: *Fluid Dynamics of Atmosphere and Oceans, Numerical Modelling, Atmospheric Physics, Oceanography, Numerical Modelling of Atmosphere and Oceans, Remote Sensing, Numerical Analysis*

MSc Mathematics (Distinction), Queen Mary University of London

2018-2020

Final Project: 'Fourier Space Methods in PDEs' (Distinction)

Key modules: *PDEs, Complex Systems, Graphs & Networks, Time Series*

BSc Mathematical Sciences (First-class honours), University of Bath

2014-2017

Academic Experience at the University of Reading

Research contributions

Organiser of the DARC (Data Assimilation Research Centre) seminar series

2023 - 2024

Organiser of SDAG (Student Data Assimilation Group)

2022 - 2024

Member of the Joint DTP Conference Organising committee, University of Surrey

2022

Multidisciplinary conference attended by around 200 PhD students.

My role included creating the conference timetable; chairing multiple sessions; organisation and running of the 'Model UN: Debating the Sustainable Development Goals' session; organising sponsors and running social activities.

Teaching experience

Vector Calculus (MA2VC) *Marking*

2024

Statistics for Weather and Climate Science (MT2SWC) *Demonstrated and marking*

2023 - 2024

School of Biological Sciences Mathematics (BI1MA17) *Demonstrated and lectured*

2023 - 2024

Chemistry Mathematics (CH1M2) *Demonstrated*

2023

Calculus (MA1CA) *Marking*

2022 - 2023

Dare to discover DA, National Centre for Earth Observation MOOC *Facilitated*

2022

Research Visits

Met Office UK, Exeter

February 2025, July 2024, October 2023, March 2023

CERFACs, Toulouse

June 2023, April 2024

Computing Skills

Languages: Proficient in python, familiar with Unix and Fortran-90

Systems: Monsoon, NEMO, NEMOVAR

Other: NetCDF files, LaTeX

Scholarships and awards

National Centre for Earth Observation Conference Best Poster Award 2023

Lloyds Scholars Programme, University of Bath 2014 - 2017

Awarded a scholarship from Lloyd's banking group. Completed a vigorous and competitive application process and 100 hours of volunteering per academic year as part of the scholarship and an internship.

Relevant training

Data Assimilation Training courses

ECMWF Data Assimilation course 2023

Dare to discover DA, National Centre for Earth Observation MOOC 2022

Data Assimilation Course, University of Reading and National Centre for Earth Observation 2022

National Centre for Earth Observation Early Careers Researcher's Forum

Data Visualisation 2024

Impact in Earth Observation 2023

The use of Artificial Intelligence and Machine Learning in Earth Observation 2022

Scientific computing courses

Introduction to Scientific Computing, Programming Design and Testing, Software Development, University of Reading 2022

Other professional experience

Private Maths Tutor 2017 - 2024

Accounts Payable Clerk, Care in Finance 2018 - 2021

Accounts Administration Assistant, Care in Finance 2017 - 2018

Financial Analyst, Executive MI & Reporting, Lloyds Banking Group 2015

Outreach

Mathematics in Weather, Oakview School November 2024

Led a daylong event targeted at Year 5-6 including presentations and activities.

Women and Girls in STEM Day, NCEO February 2024

Online interview discussing experience as a woman in STEM.

Careers Day, Chancellors School January 2024

Presented at a careers day to Year 9-10.

Mathematics at University, Ashmole Academy May 2022 and May 2023

Presentation to Year 12 about Mathematics at University including short workshop style sessions.

Presentations

Meteorology Department Lunchtime Seminar, University of Reading November 2024

Oral presentation '*Preparing for the assimilation of future ocean-current measurement*'

UKEO Conference 2024, National STEM Centre September 2024

Oral presentation '*Implications of velocity variables for variational ocean data assimilation*'

NPOP DA Meeting, Met Office June 2024

Oral presentation '*On the choice of velocity variables for variational ocean data assimilation*'

Adjoint Workshop, Lake George, NY May 2024

Oral presentation '*On the choice of velocity variables for variational ocean data assimilation*' and chaired a session

International Symposium on Data Assimilation, Bologna October 2023

Poster presentation '*On the choice of velocity variables for variational ocean data assimilation*'

NCEO Conference (Best poster award), University of Leeds September 2023

Poster presentation '*On the choice of velocity variables for variational ocean data assimilation*'

Joint DTP Conference, University of Surrey September 2023

Oral presentation '*Assimilation of future ocean-current measurements from satellites*'

ESA ATSCV Workshop, Mercator Ocean International June 2023

Oral presentation '*On the choice of velocity variables for variational ocean data assimilation*'